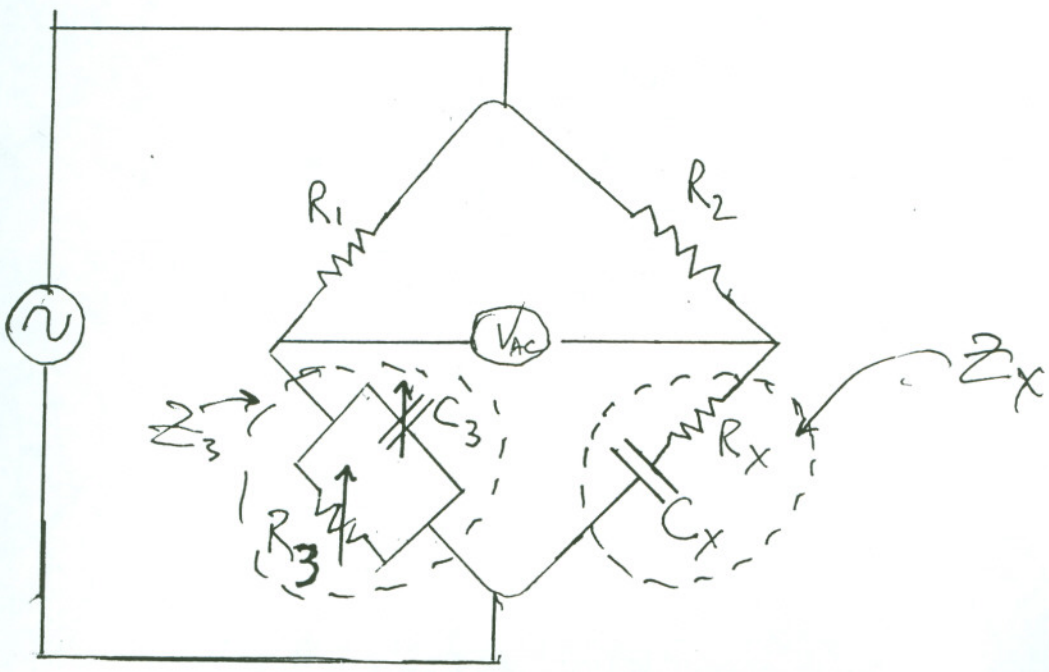
Analytical Cume **September 13, 1997**

Capacitance/conductivity data can be useful in the study of adsorption of ions and neutral species on to metal surfaces, charge changes at MOS gates in semiconductor sensors, or as a diagnostic for ion penetration through films. Historically, double layer capacitance measurements have been accomplished by either an impedance bridge, or tensametric methods. Tensametric methods are limited to mercury electrodes, leaving bridge methods as the alternative were mercury electrodes can not be used.

Shown below is an AC impedance bridge called a Wien Bridge. In general the balance condition will be given by the equation:

$$Z_x = \frac{Z_3 R_2}{R_1}$$

With complex impedances, two balance conditions must be satisfied, one for the resistive component and one for the reactive component, each orthagonal to each other. If R_1 and R_2 are fixed, there will be one set of values for R_3 and C_3 and one frequency where the null condition is met. At null, the resistive (real) parts of Zx will be equal to the real part of the right side of the above equation, and the reactive (imaginary) part of Zx will be equal to the imaginary part of the right side of the equation.

Find an equation for Rx in terms R_1, R_2, R_3, and C_3, and ω, and another equation for Cx in terms R_1, R_2, R_3, and C_3, and ω at null.